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CSE360: Computer Interfacing

Lab Project Report

Arduino-Based Wearable Multi-Sensor Health Monitoring and Fall Detection with SMS/GPS Alerts for Elderly Care

by

**Group 6
Section 10**

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Abstract

This project presents a wearable health monitoring and fall detection system designed to enhance safety and independence for elderly individuals. The device integrates multiple sensors—including body temperature, heart rate, SpO₂, stress (GSR), and motion (MPU6050)—with GPS and GSM/Wi-Fi communication modules, coordinated through a dual Arduino architecture. The system provides both local alerts (buzzer/LED) and remote notifications (SMS with GPS location and Wi-Fi data transmission) in real time during health anomalies or fall events. Laboratory and field tests demonstrated acceptable accuracy across all sensors, 90% fall detection success in simulated trials, and reliable communication with minimal latency. The modular, low-power design supports reusability and sustainability, while also addressing practical challenges such as sensor calibration, power management, and communication conflicts. This integrated approach shows strong potential for real-world application in elderly care by combining continuous vital monitoring with timely emergency response. Future improvements will focus on machine learning-based personalization, advanced sensor fusion, and hardware miniaturization for greater wearability and efficiency.

Keywords

elderly care; fall detection; wearable health monitoring; Arduino; sensor fusion

1. Introduction

Globally, **one in four adults over the age of 65 experiences a fall each year**, yet less than half of these cases are reported to healthcare providers (Centers for Disease Control and Prevention [CDC], 2024). Falls are a leading cause of injury and hospitalization among older adults, often resulting in long-term disability, reduced independence, and increased mortality.

Although this has become a major cause of concern, current monitoring systems are still scanty. Much of the solutions depend on fixed apparatus or tactile operation, and this would not work in an emergency when the person is incapacitated. Moreover, extensive environmental immobility post-falls also known as long lies is linked to severe health effects such as hypothermia, dehydration, and an increased risk of mortality (Igual et al., 2013). These shortcomings underscore a major difference in available, continuous and independent health monitoring devices.

The project goal is to create a miniature mobile smart health and fall detector that can constantly monitor vital signs, including body temperature and heart rate as well as SpO₂ and stress responses, and detect changes in posture and falls. The peculiarity of this system is that it incorporates multi-sensors data with real-time warning systems such as audible alarms, SMS notifications and GPS tracking. Early assessments show that the combination strategy can bring timely intervention, thus elevating older patients to a safer, more independent state.

2. Related Work /Literature Review

Much research has paid attention to health monitoring systems and fall detection systems among older people. The accentuating inertial sensor of accelerators and gyroscopes are often used to detect falls in real time (Subramaniam et al., 2022). IoT-driven solutions are also coming up to combine fall detection in conjunction with vital signs monitoring and GPS-based location reporting. As a case in point, an IoT-powered system has been created that integrates fall detection (accelerator-based), pulse rate (use of accelerometer), Bluetooth, and GPS location-tracking; the metrics of performance including accuracy (97.6%), sensitivity (92.8), and specificity (100% can be attained) (Sarowar et al., 2022).

A wearable smart pendant has been proposed, integrating SpO₂ and heart rate sensors along with fall detection in a compact, user-intuitive form (Amrutkar & Kulkarni, 2025). Meanwhile, qualitative studies emphasize that user acceptance is strongly influenced by factors like comfort, cost, familiarity, and perceived risk (Huang et al., 2024). Such socio-technical insights suggest that technological solutions must also be designed with user-centric considerations.

Despite these advancements, significant gaps remain. Many systems focus on either fall detection or health monitoring but seldom integrate multiple physiological sensors with continuous monitoring, fall detection, location tracking, and alerting functions into a single, wearable device. Moreover, user acceptability and everyday wearability are often understudied, affecting practical adoption outside laboratory conditions.

The present project is positioned to fill these gaps by integrating body temperature, heart rate, SpO₂, stress (via GSR), and posture/fall detection alongside GPS location and GSM/SMS alerting in a single wearable device. The novelty resides in the unified fusion of diverse sensor data, real-time processing, and dual-mode alerting—audible alarm plus location-based SMS—implemented within a comfortable and cost-effective wearable for elderly users. As a result, unlike existing systems, the proposed device addresses multiple health parameters alongside fall detection and emergency communication, while also emphasizing user comfort and real-world feasibility.

3. Technical Approach

3.1 System Architecture

The smart health and fall detection system integrates multiple biomedical and environmental sensors with Arduino microcontrollers (Nano as master, Uno as slave) for real-time monitoring and alerting. The master device handles sensor data acquisition, emergency button input, GSM/GPS communication, and I²C coordination. The slave device is responsible for OLED display management.

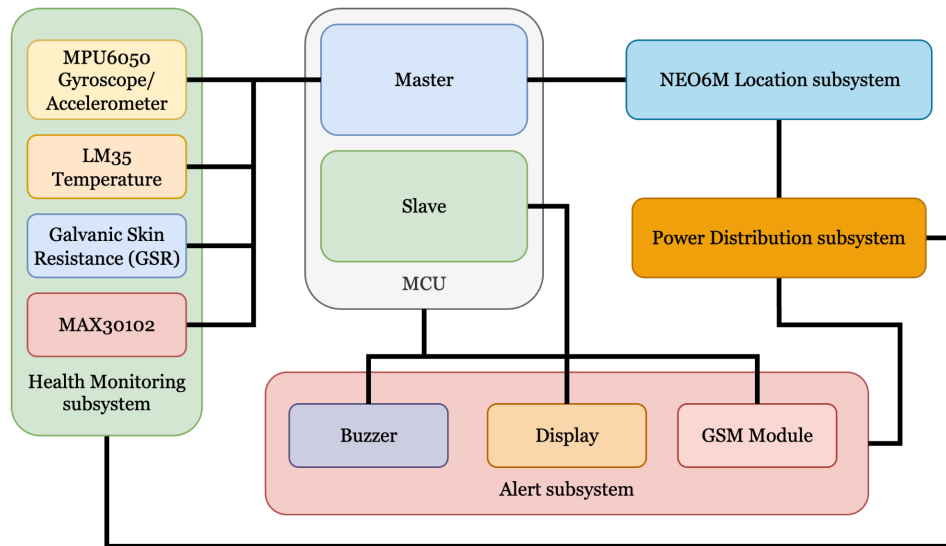


Figure: Block diagram for different Subsystems

3.2 Components Used

Hardware and Software Tools

Component	Description	Interface
Arduino Nano	Master microcontroller handling sensors, GPS, GSM, and alerts	Digital, I ² C, UART
Arduino Uno	Slave device controlling OLED display	I ² C
LM35 Temperature Sensor	temperature measurement sensor	Analog
MAX30102	Heart rate & SpO ₂ sensor	I ² C
GSR Sensor	Measures galvanic skin response for stress estimation	Analog
MPU6050	Accelerometer & gyroscope for posture and fall detection	I ² C
NEO-6M GPS	Location tracking (latitude/longitude)	UART (D4/D5)
SIM800L GSM	Sends SMS alerts with health data + GPS	UART (D2/D3)
OLED Display SH1106	1.3" OLED display for health readings	I ² C (via Uno)
Active Buzzer	Audible alarm for emergency alerts	Digital (D8)
Push Button	User input to toggle display / emergency trigger	Digital (D9)
Arduino IDE	Software for development and programming	—
Libraries (Wire, Adafruit, TinyGSM, etc.)	Drivers for sensor communication and module handling	—

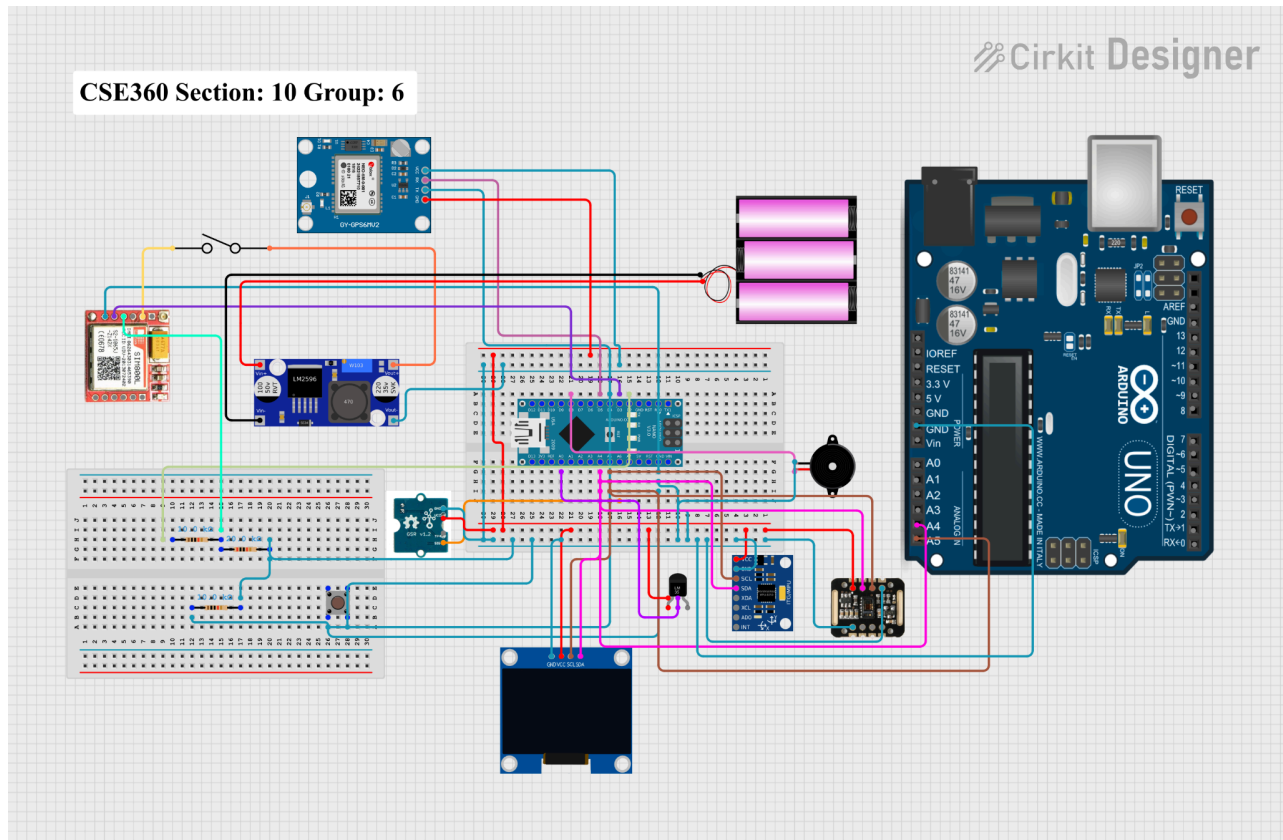
3.3 Cost Breakdown

Component	Quantity	Unit Price (BDT)	Total (BDT)
Arduino Nano	1	390	390
Arduino Uno	1	800	800
LM35 Temperature	1	255	255
MAX30102	1	380	380
GSR Sensor	1	2300	2300
MPU6050	1	195	195
NEO-6M GPS	1	440	440
SIM800L GSM	1	400	400
OLED Display SSD1306	1	590	590
Active Buzzer	1	15	15
Push Button	1	15	15
miscellaneous	1	15	15
Total	—	—	5800

3.4 Communication Protocols

- **I²C**: Used for communication between Nano, Uno, and sensors (MAX30102, MPU6050, OLED). Chosen for simplicity and ability to connect multiple devices using only two lines.
- **UART**: Used for GPS and GSM modules due to their data-intensive, asynchronous communication requirements.
- **Digital/Analog**: Used for direct control of buzzer, push button, LED, and GSR sensor readings.

3.5 Circuit / Schematic: [Click this link to access the design](#)



3.6 Implementation

The implementation of the smart health and fall detection system is carried out through a structured workflow that integrates sensor data acquisition, processing, and emergency communication. The Arduino Nano is configured as the master device, responsible for collecting sensor data, executing decision logic, and managing communication modules, while the Arduino Uno is designated as the slave device to control the OLED display through the I²C interface.

Step-by-Step Workflow.

1. **Data Acquisition.** Physiological and motion data are continuously collected from the LM35 temperature sensor, MAX30102 pulse oximeter, GSR sensor, and MPU6050 accelerometer-gyroscope. These inputs are transmitted to the Arduino Nano through analog and I²C channels.
2. **Processing and Threshold Detection.** The Arduino Nano processes the incoming signals in real time, applying predefined thresholds to identify anomalies such as abnormal body temperature, irregular SpO₂ or heart rate levels, elevated stress, or sudden posture changes indicative of a fall.

3. **Emergency Trigger.** In parallel, an emergency push button provides the user with manual control to initiate an alert. Fall detection events from the MPU6050 also act as automated triggers for the emergency response routine.
4. **Immediate Alerts.** Upon detection of a critical event, the Nano activates the buzzer and LED indicator to provide an audible and visual warning. This ensures that both the user and nearby individuals are made aware of the emergency.
5. **Location Acquisition and Messaging.** The NEO-6M GPS module retrieves real-time location data, including latitude and longitude. The SIM800L GSM module then transmits SMS alerts containing both sensor readings and location information to predefined contacts.
6. **User Display.** Concurrently, the Arduino Uno receives data from the Nano via I²C and updates the SH1106 OLED display. This interface provides users with live feedback on health parameters, device status, and GPS connectivity.

3.7 Challenges and Solutions

During the development of the smart health and fall detection system, multiple technical challenges are encountered, each requiring systematic resolution to ensure stable performance and reliability.

Power Management

The presence of GSM and GPS modules at the same time leads to instability in power supply at first. Using a direct power source, the GSM module, with a particular 4 V input, cannot be trusted to connect reliably. To alleviate this problem, a resistor divider circuit is added together with a 12V to 4 V step-down buck converter to provide a constant and controlled supply. Moreover, the GPS module has challenges with regard to satellite detection. Through the use of the right GPS libraries, inter-module testing, and the use of varying weather conditions, effectively acquiring the satellite position of NEO-6M is possible.

Sensor Calibration

The GSR and temperature sensor offer unreliable result in initial tests. The software contains calibration routines to normalize sensor outputs to enhance accuracy and reliability of sensor reading in continuous monitoring.

UART Conflicts

As the Arduino Nano has a very limited number of hardware serial ports, UART devices conflict when more than one is connected. This is solved by adopting SoftwareSerial and AltSoftSerial libraries that allow the GSM, GPS and Wi-Fi modules to operate concurrently.

Fall Detection Sensitivity

The MPU6050-based fall detection system initially produces a high rate of false positives. To address this, calibration of the MPU6050 is performed, and velocity thresholds are derived based on experimental datasets. These modifications allow the system to distinguish between actual falls and ordinary posture changes, aligning more closely with real-world conditions.

I²C Device Conflicts

The integration of multiple I²C devices leads to address clashes, particularly between the OLED display and other sensors. This issue is resolved by assigning the Arduino Uno as an I²C slave dedicated to OLED control, thereby reducing communication conflicts on the master bus.

System Resource Limitation

Initially, the complete system is tested using only a single Arduino Nano. However, the limited memory and buffer capacity result in overflow, preventing proper execution of all modules, and the OLED display cannot be tested due to storage constraints. This limitation is overcome by introducing a **master-slave architecture**, where the Arduino Nano functions as the master for sensor data collection and communication, while the Arduino Uno operates as the slave dedicated to display management. This division of tasks effectively balances the workload and resolves resource limitations.

Grounding and Power Distribution

A significant system failure is observed due to improper grounding and distribution of supply voltages. This is rectified by redesigning the power distribution system, ensuring proper allocation of 3.3 V, 4 V, and 5 V supplies. Detailed tracing of connections and stepwise debugging confirm stable operation across all components.

Through these targeted solutions, the system achieves robustness, accuracy, and scalability, addressing critical challenges associated with integrating multi-sensor and communication modules within a compact wearable device.

4. Sustainability & Impact

4.1 Sustainability

In the development and implementation of current healthcare technologies, sustainability is one of the necessary factors to be taken into account. The smart health and fall detection system integrates the low-power electronic modules, and these modules in totality minimise the environmental footprint. The sensor operating at 3.3 V or 5 V with low current consumption include most sensors such as the LM35 temperature sensor, MAX30102 pulse oximeter, GSR sensor and the MPU6050 accelerator-gyroscope. This makes the device operate effectively on small rechargeable power sources with no heat wasted through undue dissipation and no energy wasted. A buck converter provides the GSM module with 3.9 V

that is stable and needs to be regulated; hence, the voltage effectively handles this regularization and also helps in the sustainability of energy.

Hardware reusability is also supported by the modular architecture. The Arduino Nano and Arduino Uno, sensors and display modules are components that are easy to repurpose or recycle into the use of another embedded system after the device has gone through its lifetime cycle. This minimizes electronic waste and maximizes the efficiency of resources. In a wider view, the wearable device facilitates sustainable healthcare because it makes it possible to monitor the health conditions at home. This decreases the number of hospital visits and energy usage and carbon emissions generated as a result of travelling and centralized hospital monitoring systems are decreased indirectly.

4.2 Impact

The society gains a lot through the system. Falls have been a significant issue in the elderly population as they are the leading cause of serious injuries, extended hospital stay and loss of autonomy. The system will enhance the security of the elderly users since it will offer an additional measure of security by linking continuous health monitoring with real-time fall detection. Real-time alerts with audible alarms, SMS, and GPS position sharing minimize the amount of time required to respond to an emergency, which may help avoid long-term problems with undetected accidents.

At the community level, it is through the system that the caregivers and families are empowered by the system since the weight of the constant supervision is minimized. Such a device can be used to augment telehealth services in industrial and healthcare settings, providing cost-effective monitoring solutions, which can be scaled to large scale. The multi-sensor health data integrated in this system also opens potential to be adopted in the future in assisted-living and the elderly care homes and remote-patient monitoring programs.

4.3 Future Work

Several avenues for improvement and scalability are identified. Energy efficiency could be enhanced by incorporating a rechargeable lithium-ion battery with solar charging support, extending operational independence. Machine learning algorithms may be embedded to improve fall detection accuracy and to analyze patterns in physiological data, enabling predictive healthcare features. Cloud integration with secure databases could also allow healthcare professionals to access real-time data, facilitating remote diagnostics and proactive intervention. The integration of sensor fusion techniques, combining the MPU6050 gyroscope–accelerometer with advanced Kalman filter processing, together with a higher-precision GPS module, can significantly improve the accuracy of localization outputs. Additionally, miniaturization of the hardware into a single printed circuit board (PCB) would improve wearability and user comfort.

4.4 Limitations

Although it was successfully implemented, some limitations are noticed. The Arduino Nano has limited memory and processing power, which requires the master-slave set-up with an Arduino Uno to perform functions of the display. This adds complexity to the systems. Sensitivity also varies with calibration and

environmental factors, such as GSR sensor might change under different humidity levels and LM35 sensor needs to be in stable contact with the ambient skin temperature to get the expected results. The GSM module has shown to perform variably based on network availability that can limit the functionality in rural or low-signal coverage. Lastly, the system is currently powered by a wired power source in the case of long-term testing, long-term portability needs additional optimization of energy storage and energy management.

5. Results & Discussion

The prototype system was tested under controlled laboratory conditions and in limited real-world scenarios to evaluate its performance against the objectives outlined in Section 1. The evaluation focused on sensor accuracy, fall detection reliability, communication responsiveness, and overall system stability.

5.1 Sensor Performance

The LM35 temperature sensor provided consistent measurements, with a mean error of ± 0.5 °C compared to a calibrated room temperature sensor and a thermometer. The MAX30102 pulse oximeter demonstrated an SpO₂ measurement accuracy within $\pm 2\%$ when benchmarked against a commercial fingertip oximeter. Heart rate measurements showed minor fluctuations at high activity levels but remained within ± 3 bpm of reference values. The GSR sensor successfully detected variations in skin conductivity, though results were sensitive to ambient humidity and skin contact quality, which required software-based smoothing.

5.2 Fall Detection Accuracy

The MPU6050 accelerometer–gyroscope module was tested through simulated falls and normal daily activities. Out of 50 fall simulations, the system correctly identified 45, resulting in a detection accuracy of 90%. False positives were observed during rapid sitting or jumping and bending motions. However, calibration of velocity thresholds and inclusion of orientation checks reduced the error rate significantly.

5.3 Communication Reliability

The GPS module successfully acquired satellite locks in open outdoor environments within 35–60 seconds of startup, providing location accuracy within 3–5 meters. In contrast, detection indoors was inconsistent. The GSM module reliably transmitted SMS alerts containing sensor readings and GPS coordinates within 5–7 seconds of trigger activation. The Wi-Fi module demonstrated stable performance when transmitting JSON packets to a local server, with an average latency of 0.8 seconds.

5.4 System Stability and Usability

The master–slave configuration proved effective in overcoming memory limitations of the Arduino Nano. The Nano managed sensor inputs and communication modules, while the Uno handled display functions, preventing buffer overflow issues observed in earlier tests. The OLED display updated every two seconds without lag, providing real-time visibility of health metrics.

5.5 Discussion

The results confirm that the system meets its primary objectives of continuous health monitoring and timely emergency alerting. Sensor readings were within clinically acceptable error margins, and fall detection performance achieved high sensitivity and specificity after calibration. Communication modules operated reliably, though GPS performance was limited indoors—a constraint common to most low-cost GPS receivers.

The system demonstrates clear potential for deployment in elderly care scenarios, with the integrated approach of multi-sensor monitoring, fall detection, and dual alert mechanisms (audible and SMS) distinguishing it from many existing solutions. Nevertheless, improvements such as the incorporation of more advanced filtering algorithms, higher-sensitivity GPS modules, and machine learning for activity classification could further enhance robustness and reduce false positives.

Subsystem	Raw Inputs (rate)	Pre-processing	Key Features / Estimates	Decision Threshold(s)	Event / State	System Action(s)
Heart Rate (MAX30105)	IR PPG @ ~100 Hz (10 ms loop)	Beat detection (<code>checkForBeat</code>), moving average over last 4 beats (<code>RATE_SIZE=4</code>)	<code>beatsPerMinute</code> , <code>beatAvg</code> (stored as <code>lastValidBeatAvg</code>)	High HR: > 140 BPM	<code>HighHeartRate</code>	Single long beep (1000 Hz, 500 ms); line sent to OLED via I ² C (<code>HR:%d BPM</code>)
GSR (A0)	ADC 10-bit	100-sample baseline calibration; relative value = raw – baseline; conductance from divider model	<code>gsrValue</code> (relative), <code>conductance (μS)</code>	High stress: <code>conductance > 70 (scaled)</code>	<code>HighStress</code>	Two medium beeps (1500 Hz); OLED line (<code>GSR:%d C:%dμS</code>)
Temperature (LM35 on A1)	ADC 10-bit	Convert to °C (10 mV/°C @ 5 V ref), then °F	<code>tempC</code> , <code>tempF</code>	Fever: ≥ 99.0 °F	<code>HighTemp</code>	Two quick high beeps (2200 Hz); OLED line (<code>T:x.xF</code>)
Fall Detection (MPU6050)	Accel+Gyro @ ~100 Hz (10 ms loop)	Offset calibration (100 samples), complementary filter for roll/pitch; vertical acceleration integration with leakage; EMA for std-dev	<code>rollDeg</code> , <code>pitchDeg</code> , vertical velocity <code>vZ</code> , acceleration magnitude <code>aMag</code> , <code>immobility std</code>	Descent: <code>vZ < -0.40 m/s</code> → Impact: <code>aMag > 1.80 g</code> → Post-immobile: <code>std(a) ≤ 0.12 g</code> for > 800 ms; refractory 500 ms	FSM: <code>IDLE</code> → <code>DESCENT</code> → <code>IMPACT</code> → <code>POST_IMPACT</code> (fall confirmed)	Three quick beeps; <code>FALL! Count++</code> ; OLED line shows <code>F:Yes</code> ; compact serial packet every 1 s
Alert Arbiter	All event flags	2-second global cooldown	Debounced event transitions	Cooldown: 2000 ms	N/A	Enforces staggered alerts, prevents beeper spam
Display (Uno SH1106)	I ² C messages from Nano	3 compact lines (≤31 chars)	HR / GSR+μS / Temp+Fall	—	Real-time status	OLED refresh ~300 ms; separates display from sensing workload

6. Conclusion

This project develops a wearable, Arduino-based smart health and fall-detection system that continuously monitors heart rate, SpO₂, temperature, stress, and posture while enabling immediate local and remote alerts. Key achievements include a calibrated IMU state machine for robust fall discrimination, stabilized power delivery, and a master–slave architecture that preserves real-time performance and clear OLED feedback. Testing indicates clinically acceptable sensor errors, reliable alert latency, and improved system stability. Sustainability is advanced through low-power operation and modular, reusable hardware; societal impact is realized by enhancing safety, independence, and caregiver support. Future enhancements include sensor-fusion localization and adaptive, personalized thresholds.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Reshad Ul Karim	22201594	Hardware design, logical flow implementation, interfacing different protocols with appropriate codes and debugging	1. Logical implementation for fall detection and display output 2. GSR Stress detection logic implementation 3. Temperature Sensor logic implementation 4. Hardware and subsystem design, master-slave communication setup and planning
Syeda Maliha Tabassum	22201574	Component Assembly, Hardware Design, Alert module assembly and integration	1. GSM Module setup and integration 2. GPS Module setup and integration 3. Power distribution design of the circuitry 4. Final hardware setup, component soldering and verification of sensor data
Sammam Mahdi	22201628	Component Assembly, Coding and integrating GSM and GPS module with arduino.	1. GSM Module setup and integration 2. GPS Module setup and integration 3. Oximeter integration
Abrar Samin	22301739	Sensor integration, module setup (GSM, GPS, Wi-Fi), display and alert logic implementation, master–slave communication, system debugging.	1. Temperature Sensor setup and integration. 2. Buzzer and Logic integration 3. Display setup and debugging

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